THE COHOMOLOGY SHEAVES OF THE NAIVE COTANGENT COMPLEX OF A MORPHISM OF ALGEBRAIC SPACES ARE QUASI-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5397-5403 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 5405-5423, the complete author proof (17 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The naive cotangent complex of a morphism, conormal sheaves and differentials of a presentation, and quasi-coherence of the resulting sheaves; the identification is retained.

Source reference. Original lemma lemma-NL-compare-spaces-schemes: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5324–5334. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-NL-etale-localization: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5297–5308. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-module-differentials-quasi-coherent: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1017–1020. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-netherlander-quasi-coherent: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5423–5423. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma section-sheaf-differentials: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 839–877. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma sites-modules-section-netherlander: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5423–5423. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-properties-lemma-characterize-quasi-coherent: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5423–5423. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The cohomology sheaves of the complex $NL_{X/Y}$ are quasi-coherent, zero outside degrees -1, 0 and equal to $\Omega_{X/Y}$ in degree 0.*

Proof. By construction of the naive cotangent complex in Modules on Sites, Section sites-modules-section-netherlander we have that $NL_{X/Y}$ is a complex sitting in degrees -1, 0 and that its cohomology in degree 0 is $\Omega_{X/Y}$ (by our construction of $\Omega_{X/Y}$ in Section section-sheaf-differentials). The sheaf of differentials is quasi-coherent (by Lemma lemma-module-differentials-quasi-coherent). To finish the proof it suffices to show that $H^{-1}(NL_{X/Y})$ is quasi-coherent. This follows by checking étale locally (allowed by Lemma lemma-NL-etale-localization and Properties of Spaces, Lemma spaces-properties-lemma-characterize-quasi-coherent) reducing to the case of schemes (Lemma lemma-NL-compare-spaces-schemes) and finally using the result in the case of schemes (More on Morphisms, Lemma more-morphisms-lemma-netherlander-quasi-coherent). □